THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE UNIT TEST SET B

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon, and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Producers in a food chain are: 1

(a) green plants **(b)** herbivores **(c)** decomposers **(d)** carnivores

2. Iodine solution turns blue-black in the presence of: 1

(a) water **(b)** starch **(c)** protein **(d)** glucose

3. Plants grown in the dark become: 1

(a) full of starch **(b)** dark green

(c) pale and weak **(d)** taller and stronger

4. Stomata are surrounded by: 1

(a) root cells **(b)** guard cells **(c)** bark cells **(d)** phloem cells

5. In a kaleidoscope, mirrors are placed at: 1

(a) far apart parallel **(b)** flat on table

(c) perpendicular to the Sun **(d)** an angle to each other

6. An object that produces its own light is called: 1

(a) transparent **(b)** opaque **(c)** luminous **(d)** non-luminous

7. Frosted glass is: 1

(a) translucent **(b)** transparent **(c)** opaque **(d)** luminous

8. Solar eclipse shadows are formed because the Moon is: 1

(a) luminous **(b)** translucent **(c)** transparent **(d)** opaque

9. The spinning of the Earth on its axis is called: 1

(a) eclipse **(b)** rotation **(c)** orbit **(d)** revolution

10. A lunar eclipse can be seen: 1

(a) from anywhere on the night side of Earth **(b)** nowhere

(c) only at poles **(d)** only at the equator

11. The Moon is a: 1

(a) star **(b)** natural satellite

(c) planet **(d)** comet

12. The Earth takes about ___ to revolve once around the Sun. 1

(a) 24 hours **(b)** 365¼ days **(c)** 30 days **(d)** 27 days

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 15–25 words each.

13. Name the tissues that transport water and food. 2

14. Write the equation of photosynthesis in words. 2

15. State two characteristics of the image in a plane mirror. 2

16. Why are there seasons on Earth? 2

17. What are solstices? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Why do variegated leaves help in studying photosynthesis? 3

19. Compare xylem and phloem. 3

20. Why do we see a clear image in a mirror but not on a wall? 3

21. How is a leap year calculated? Give two examples. 3

22. How does the size of a shadow change with the distance of the object from the source? 3

OR

Describe an activity to show that light travels in a straight line. 3

23. Explain the formation of a lunar eclipse. 3

OR

Describe the phases of the Moon in a lunar month. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

24. **Case study:** A student covers a part of a leaf with black paper for two days, then tests it for starch. 5

(i) Aim?
(ii) Why remove chlorophyll?
(iii) Observation?
(iv) Conclusion?

OR

(i) What is transpiration? 5
(ii) Factors affecting it.
(iii) Its benefits.

25. **Case study:** A solar eclipse is predicted. 5

(i) Which day will it occur?
(ii) Which bodies line up?
(iii) How to watch safely?
(iv) Why not look directly?

OR

(i) Why do eclipses not happen every month? 5
(ii) How often do they occur?
(iii) Use of eclipse studies.

END OF QUESTION PAPER — Total: 50 Marks

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